Reproducibility - Paper X

Everything in this paper except the live-model experiment is deterministic and reproducible offline in under a minute. This file says exactly which command reproduces which claim, and - as honestly - which claim **cannot** yet be reproduced because the experiment has not been run.

Environment

- Python 3.11 (tested 3.11.15). Dependencies and tested versions: `requirements.txt`.
- `pip install -r requirements.txt` (numpy, scipy, matplotlib; pytest for tests).
- Or use the container: `docker build -t paperx . && docker run --rm paperx`.

One command

```
make all        # install + verify (harness) + test-all (both suites)
```

What reproduces what

Claim in the paper	Command	Artefact produced	What it establishes
Theorems 1-6 numerically consistent with their closed forms (E1-E9)	`make verify` (`code/experiment_coscaling.py`)	`results/verdicts.json`, `results/report.txt`, `figures/*.png`	code matches the maths (10/10 internal-consistency + integrator checks)
Regression: harness output is stable	`make test` (`code/test_coscaling.py`)	-	12 assertions on the harness
Independent re-derivation of the corrected theorems	`make test-independent` (`code/test_theorems_independent.py`)	-	14 from-scratch checks (no harness import): Thm 2 time-varying limits, Thm 4 equality = linear divergence, Thm 5 non-normal transient + null-axis floor, Thm 6 OU variance, $\gamma_3=1$ boundary
β/k estimator recovers known exponents	`make estimator`	stdout	synthetic identifiability only
Real-model harness plumbing	`make selftest`	`results/realmodel/*_selftest.json`	deterministic stub, **not data**

The two test suites are deliberately independent: `test_coscaling.py` shares the harness integrator (regression), while `test_theorems_independent.py` re-implements the dynamics from scratch with `scipy.solve_ivp` + matrix exponentials, so a shared bug cannot hide.

Determinism

Random seed is fixed (`7`) in the harness; figures and verdicts are byte-reproducible on a fixed library stack. `code/experiment_coscaling.py` writes to the current directory by default (`COSCALING_OUTDIR=.`); run it from the paper root to regenerate the canonical `results/` and `figures/`.

What is NOT reproducible here (stated plainly)

- The **`β > k` threshold on a real system.** Not run. The decisive experiment is pre-registered in `experiments/PROTOCOL.md`; reproducing it needs live API keys for several model families and a network-isolated sandbox (`SECURITY.md`).
- The **first Claude run** is a pilot: `n = 1`, one task, same-family scorer, IV.d non-compliant, H1/H2 false. It corroborates the corrector *mechanism* only and is **not** evidence for the threshold (see `NEGATIVE_RESULTS.md`).

Continuous integration (optional)

CI is not wired up as a live GitHub Actions workflow here (the repository's Actions policy must first allow the GitHub-authored `actions/checkout` and `actions/setup-python`). The exact verification commands are the `Makefile` targets, so a one-job workflow is just:

```
# .github/workflows/paperx-ci.yml  (enable once Actions allows GitHub-authored actions)
on: [push, pull_request]
jobs:
  verify:
    runs-on: ubuntu-latest
    defaults: { run: { working-directory: papers/Paper-X-Coupled-CoScaling-Correction } }
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with: { python-version: '3.11' }
      - run: pip install -r requirements.txt
      - run: cd code && python experiment_coscaling.py && python -m pytest -q
      - run: cd experiments/scripts && python estimate_exponents.py --selftest
```

Until then, `make all` (or the `Dockerfile`) is the canonical, environment-independent verification path.

Honesty ladder

The status of every load-bearing claim - proved-in-model / internally-verified / synthetically-validated / real-model-pilot / open - is in `CLAIMS.md`. Nothing in this repository should be read one rung higher than it is placed there.